EXPO
engineering ▲ design

TerraCrate

Secure, offline-first file storage for local networks

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Value Proposition

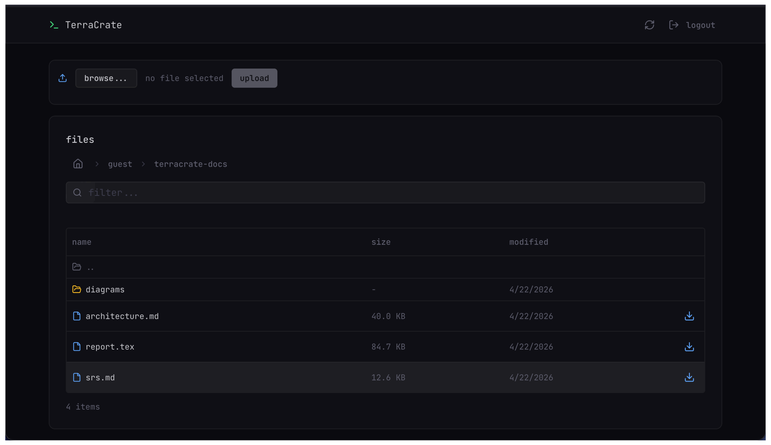
No lightweight solution combines certificate-based authentication, fine-grained access control, and zero-config discovery in one device.

Objective

Build a portable Raspberry Pi file server with mTLS authentication, flexible permissions, and mDNS discovery. Deployable anywhere without cloud infrastructure or client software beyond a browser.

Problem Definition

There is no lightweight, portable and secure file sharing solution that provides multi-user access without external dependencies.

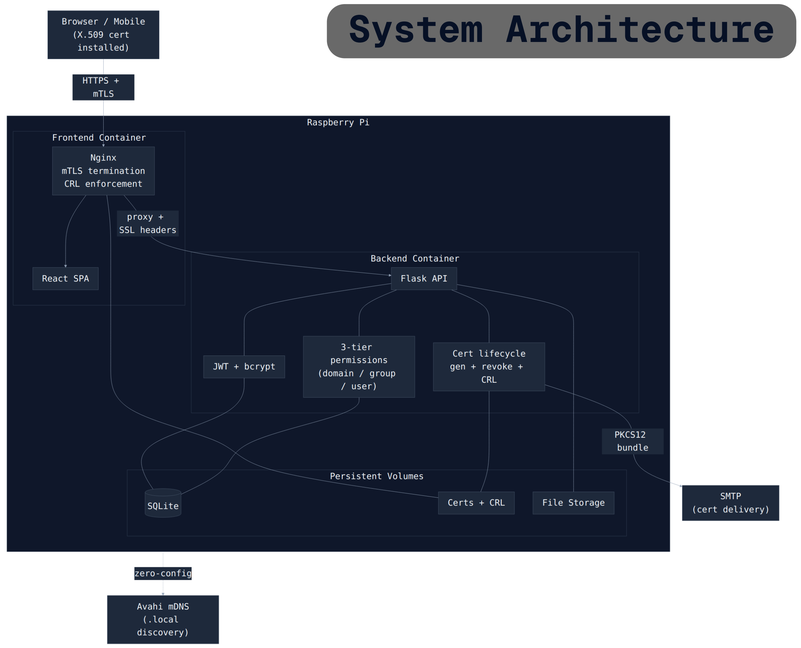


Users can browse shared files directly in the browser after installing their client certificate.

Key Requirements

- Clients must authenticate with mTLS
- Full certificate lifecycle managed on-device
- Flexible access control
- mDNS for zero-config discovery
- Fully auditable self contained system.

System Architecture



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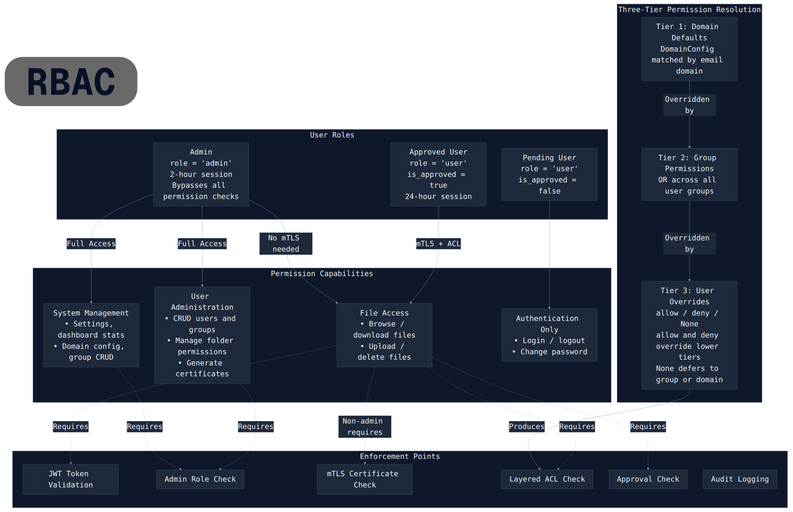
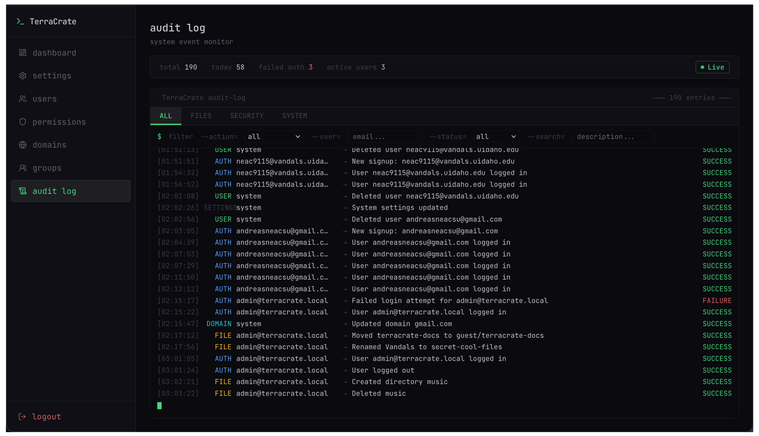


Concept Development

Two prior designs were evaluated: an SMB-over-mTLS approach requiring client-side Docker, and a QR code JWT approach lacking mutual TLS and per-user access control. The final design resolves both by delegating mTLS verification to Nginx, requiring only a browser with an installed certificate.

Validation

- 105 backend unit tests: auth, permissions, certificate lifecycle, path traversal
- Frontend enforced via ESLint, TypeScript strict mode, and Vite builds
- Three-workflow GitHub Actions CI: lint, type-check, tests, and Docker builds on every push



The real-time audit log captures every security-relevant action with timestamp, user, and outcome for auditability.

Conclusion

TerraCrate delivers a secure, portable, self-hosted file sharing system with no cloud dependencies. mTLS verification delegated to Nginx provides cryptographic mutual authentication requiring only a browser. The three-tier permissions, certificate lifecycle, mDNS discovery, and audit logging satisfy all core requirements.

Acknowledgements

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